

Trabajo Práctico N° 2: Optimización.

Ejercicio 1.

Calcular la distancia mínima del punto $(0, 4)$ a la recta $4t - 3x = 13$.

$$3x = 4t - 13$$

$$x = \frac{4t-13}{3}$$

$$x = \frac{4}{3}t - \frac{13}{3}.$$

$$\phi(t) = \frac{4}{3}t - \frac{13}{3}.$$

La longitud de las curvas $x(t)$ está dada por:

$$L = \int_0^{t_1} \sqrt{1 + [x'(t)]^2} dt,$$

desde $P(0, 4)$ hasta $\phi(t) = \frac{4}{3}t - \frac{13}{3}$.

$$F(t, x, x') = \sqrt{1 + [x'(t)]^2}.$$

Condición de Euler:

$$F_x - \frac{d}{dt} F_{x'} = 0$$

$$0 - \frac{d}{dt} \frac{1}{2} \frac{1}{\sqrt{1+[x'(t)]^2}} 2x'(t) = 0$$

$$\frac{d}{dt} \frac{x'(t)}{\sqrt{1+[x'(t)]^2}} = 0$$

$$\frac{x'(t)}{\sqrt{1+[x'(t)]^2}} = k$$

$$\sqrt{1 + [x'(t)]^2} = \frac{x'(t)}{k}$$

$$1 + [x'(t)]^2 = \left[\frac{x'(t)}{k}\right]^2$$

$$1 + [x'(t)]^2 = \frac{[x'(t)]^2}{k^2}$$

$$k^2 \{1 + [x'(t)]^2\} = [x'(t)]^2$$

$$k^2 + k^2[x'(t)]^2 = [x'(t)]^2$$

$$[x'(t)]^2 - k^2[x'(t)]^2 = k^2$$

$$(1 - k^2) [x'(t)]^2 = k^2$$

$$[x'(t)]^2 = \frac{k^2}{1-k^2}$$

$$x'(t) = \sqrt{\frac{k^2}{1-k^2}}$$

$$x'(t) = \frac{k}{\sqrt{1-k^2}}$$

$$x'(t) = c$$

$$\int x'(t) dt = \int c dt$$

$$x^*(t) = ct + A.$$

Condiciones inicial y final:

$$\begin{aligned} x^*(0) &= 4 \\ c \cdot 0 + A &= 4 \\ 0 + A &= 4 \\ A &= 4. \end{aligned}$$

$$x^*(t) = ct + 4.$$

$$\begin{aligned} x^*(t_1) &= \phi(t_1) \\ ct_1 + 4 &= \frac{4}{3}t_1 - \frac{13}{3} \\ \frac{4}{3}t_1 - ct_1 &= 4 + \frac{13}{3} \\ t_1\left(\frac{4}{3} - c\right) &= \frac{25}{3} \\ t_1\left(\frac{4-3c}{3}\right) &= \frac{25}{3} \\ t_1 &= \frac{\frac{25}{3}}{\frac{4-3c}{3}} \\ t_1 &= \frac{25}{4-3c}. \end{aligned}$$

Condición de transversalidad:

$$\begin{aligned} \{F + F_{x'}[\phi'(t) - x'(t)]\} \big|_{t=t_1} &= 0 \\ \{\sqrt{1 + [x'(t)]^2} + \frac{x'(t)}{\sqrt{1 + [x'(t)]^2}}[\phi'(t) - x'(t)]\} \big|_{t=t_1} &= 0 \\ \sqrt{1 + [x'(t_1)]^2} + \frac{x'(t_1)}{\sqrt{1 + [x'(t_1)]^2}}[\phi'(t_1) - x'(t_1)] &= 0 \\ \sqrt{1 + c^2} + \frac{c}{\sqrt{1 + c^2}}\left(\frac{4}{3} - c\right) &= 0 \\ \sqrt{1 + c^2} + \frac{c}{\sqrt{1 + c^2}}\frac{4-3c}{3} &= 0 \\ \frac{c}{\sqrt{1 + c^2}}\frac{4-3c}{3} &= -\sqrt{1 + c^2} \\ 4c - 3c^2 &= -3\sqrt{1 + c^2}\sqrt{1 + c^2} \\ 4c - 3c^2 &= -3(1 + c^2) \\ 4c - 3c^2 &= -3 - 3c^2 \\ 4c &= -3 - 3c^2 + 3c^2 \\ 4c &= -3 \\ c &= \frac{-3}{4}. \end{aligned}$$

$$\begin{aligned} t_1 &= \frac{25}{4-3\left(\frac{-3}{4}\right)} \\ t_1 &= \frac{25}{4+\frac{9}{4}} \\ t_1 &= \frac{25}{\frac{25}{4}} \\ t_1 &= 4. \end{aligned}$$

$$x^*(t) = \frac{-3}{4}t + 4, \quad \text{con } t \in [0, 4].$$

Extremal del funcional.

Condición de Legendre:

$$F_{x'x'} = \frac{\sqrt{1+[x'(t)]^2} - x'(t) \frac{x'(t)}{\sqrt{1+[x'(t)]^2}}}{[\sqrt{1+[x'(t)]^2}]^2}$$

$$F_{x'x'} = \frac{\sqrt{1+[x'(t)]^2} - \frac{[x'(t)]^2}{\sqrt{1+[x'(t)]^2}}}{1+[x'(t)]^2}$$

$$F_{x'x'} = \frac{\frac{1+[x'(t)]^2 - [x'(t)]^2}{\sqrt{1+[x'(t)]^2}}}{1+[x'(t)]^2}$$

$$F_{x'x'} = \frac{1}{\{1+[x'(t)]^2\}^{\frac{3}{2}}} > 0, \forall t.$$

Entonces, se tiene:

$$L = \int_0^4 \sqrt{1 + \left(\frac{-3}{4}\right)^2} dt$$

$$L = \int_0^4 \sqrt{1 + \frac{9}{16}} dt$$

$$L = \int_0^4 \sqrt{\frac{25}{16}} dt$$

$$L = \int_0^4 \frac{5}{4} dt$$

$$L = \left[\left(\frac{5}{4} t\right) \Big|_0^4 \right]$$

$$L = \frac{5}{4} [(t) \Big|_0^4]$$

$$L = \frac{5}{4} (4 - 0)$$

$$L = \frac{5}{4} * 4$$

$$L = 5.$$

Por lo tanto, la distancia mínima del punto a la recta es 5.

Ejercicio 2.

Maximizar el siguiente funcional con $x(0) = 1$ y $x(1) = 2$:

$$\int_0^1 [1 - x^2 - (x')^2] dt.$$

$$F(t, x, x') = 1 - x^2 - (x')^2.$$

Condición de Euler:

$$F_x - \frac{d}{dt} F_{x'} = 0$$

$$-2x - \frac{d}{dt} (-2x') = 0$$

$$-2x - (-2x'') = 0$$

$$-2x + 2x'' = 0$$

$$2(x'' - x) = 0$$

$$x'' - x = \frac{0}{2}$$

$$x'' - x = 0.$$

$$P(\lambda) = \lambda^2 - 1 \quad \Rightarrow \quad \lambda_1 = 1; \quad \lambda_2 = -1.$$

$$x^*(t) = C_1 e^t + C_2 e^{-t}.$$

Condiciones inicial y final:

$$x^*(0) = 1$$

$$C_1 e^0 + C_2 e^{-0} = 1$$

$$C_1 * 1 + C_2 * 1 = 1$$

$$C_1 + C_2 = 1$$

$$C_2 = 1 - C_1.$$

$$x^*(1) = 2$$

$$C_1 e^1 + C_2 e^{-1} = 2$$

$$C_1 e + (1 - C_1) e^{-1} = 2$$

$$C_1 e + e^{-1} - C_1 e^{-1} = 2$$

$$C_1 (e - e^{-1}) = 2 - e^{-1}$$

$$C_1 = \frac{2 - e^{-1}}{e - e^{-1}}$$

$$C_1 = \frac{2 - \frac{1}{e}}{e - \frac{1}{e}}$$

$$C_1 = \frac{\frac{2e - 1}{e}}{\frac{e^2 - 1}{e}}$$

$$C_1 = \frac{2e - 1}{e^2 - 1}.$$

$$C_2 = 1 - \frac{2e - 1}{e^2 - 1}$$

$$C_2 = \frac{e^2 - 1 - 2e + 1}{e^2 - 1}$$

$$C_2 = \frac{e^2 - 2e}{e^2 - 1}.$$

$$x^*(t) = \frac{2e-1}{e^2-1} e^t + \frac{e^2-2e}{e^2-1} e^{-t}, \quad \text{con } t \in [0, 1]. \quad \text{Extremal del funcional.}$$

Condición de Legendre (necesaria):

$$F_{x'x'} = -2 < 0.$$

Condición del Hessiano (suficiente):

$$|H F(t, x, x')| = \begin{vmatrix} F_{xx} & F_{xx'} \\ F_{x'x} & F_{x'x'} \end{vmatrix}$$

$$|H F(t, x, x')| = \begin{vmatrix} -2 & 0 \\ 0 & -2 \end{vmatrix},$$

donde:

$$|H_1| = -2 < 0 \text{ y } |H_2| = 4 > 0.$$

Por lo tanto, $x^*(t)$ es el extremal del funcional y realiza el máximo, ya que la matriz $H F(t, x, x')$ es definida negativa $\forall t, x, x' \in Dom_F$, ya que $(-1)^j |H_j| > 0$, $j = 1, 2$, y, entonces, $F(t, x, x')$ es estrictamente cóncava, $\forall t, x, x' \in Dom_F$.

Ejercicio 3.

Minimizar el siguiente funcional con $c > 0$, $x(0) = x_0$ y $x(T) = 0$.

$$\int_0^T [x^2 + c^2(x')^2] dt.$$

$$F(t, x, x') = x^2 + c^2(x')^2.$$

Condición de Euler:

$$F_x - \frac{d}{dt} F_{x'} = 0$$

$$2x - \frac{d}{dt} 2c^2 x' = 0$$

$$2x - 2c^2 x'' = 0$$

$$-2(c^2 x'' - x) = 0$$

$$c^2 x'' - x = \frac{0}{-2}$$

$$c^2 x'' - x = 0.$$

$$P(\lambda) = c^2 \lambda^2 - 1 \quad \Rightarrow \quad \lambda_1 = \frac{1}{c}; \quad \lambda_2 = \frac{-1}{c}.$$

$$x^*(t) = C_1 e^{\frac{1}{c}t} + C_2 e^{\frac{-1}{c}t}.$$

Condiciones inicial y final:

$$x^*(0) = x_0$$

$$C_1 e^{\frac{1}{c} \cdot 0} + C_2 e^{\frac{-1}{c} \cdot 0} = x_0$$

$$C_1 e^0 + C_2 e^0 = x_0$$

$$C_1 \cdot 1 + C_2 \cdot 1 = x_0$$

$$C_1 + C_2 = x_0$$

$$C_2 = x_0 - C_1.$$

$$x^*(T) = 0$$

$$C_1 e^{\frac{1}{c}T} + C_2 e^{\frac{-1}{c}T} = 0$$

$$C_1 e^{\frac{1}{c}T} + (x_0 - C_1) e^{\frac{-1}{c}T} = 0$$

$$C_1 e^{\frac{1}{c}T} + x_0 e^{\frac{-1}{c}T} - C_1 e^{\frac{-1}{c}T} = 0$$

$$C_1 (e^{\frac{1}{c}T} - e^{\frac{-1}{c}T}) = 0 - x_0 e^{\frac{-1}{c}T}$$

$$C_1 = \frac{-x_0 e^{\frac{-1}{c}T}}{e^{\frac{1}{c}T} - e^{\frac{-1}{c}T}}$$

$$C_1 = \frac{-x_0 e^{\frac{-1}{c}T}}{e^{\frac{1}{c}T} - e^{\frac{-1}{c}T}}$$

$$C_1 = \frac{\frac{-x_0}{e^{\frac{1}{c}T}}}{\frac{e^{\frac{1}{c}T} - 1}{e^{\frac{1}{c}T}}}$$

$$C_1 = \frac{-x_0}{e^{\frac{2}{c}T} - 1}.$$

$$C_2 = x_0 - \left(\frac{-x_0}{e^{\frac{2}{c}T} - 1} \right)$$

$$C_2 = x_0 + \frac{x_0}{e^{\frac{2}{c}T} - 1}$$

$$C_2 = \frac{x_0 e^{\frac{2}{c}T} - x_0 + x_0}{e^{\frac{2}{c}T} - 1}$$

$$C_2 = \frac{x_0 e^{\frac{2}{c}T}}{e^{\frac{2}{c}T} - 1}.$$

$$x^*(t) = \frac{-x_0}{e^{\frac{2}{c}T} - 1} e^{\frac{1}{c}t} + \frac{x_0 e^{\frac{2}{c}T}}{e^{\frac{2}{c}T} - 1} e^{-\frac{1}{c}t}, \quad \text{con } t \in [0, T]. \text{ Extremal del funcional.}$$

Condición de Legendre (necesaria):

$$F_{x'x'} = 2 > 0.$$

Condición del Hessiano (suficiente):

$$|H F(t, x, x')| = \begin{vmatrix} F_{xx} & F_{xx'} \\ F_{x'x} & F_{x'x'} \end{vmatrix}$$

$$|H F(t, x, x')| = \begin{vmatrix} 2 & 0 \\ 0 & 2c^2 \end{vmatrix},$$

donde:

$$|H_1| = 2 > 0 \text{ y } |H_2| = 4c^2 > 0.$$

Por lo tanto, $x^*(t)$ es el extremal del funcional y realiza el mínimo, ya que la matriz $H F(t, x, x')$ es definida positiva $\forall t, x, x' \in Dom_F$, ya que $|H_j| > 0, j = 1, 2$, y, entonces, $F(t, x, x')$ es estrictamente convexa, $\forall t, x, x' \in Dom_F$.

Ejercicio 4.

Minimizar el siguiente funcional con $x(0) = 1$:

$$\int_0^T [x^2 + (x')^2] dt.$$

$$F(t, x, x') = x^2 + (x')^2.$$

Condición de Euler:

$$F_x - \frac{d}{dt} F_{x'} = 0$$

$$2x - \frac{d}{dt} (2x') = 0$$

$$2x - (2x'') = 0$$

$$2x - 2x'' = 0$$

$$-2(x'' - x) = 0$$

$$x'' - x = \frac{0}{-2}$$

$$x'' - x = 0.$$

$$P(\lambda) = \lambda^2 - 1 \quad \Rightarrow \quad \lambda_1 = 1; \quad \lambda_2 = -1.$$

$$x^*(t) = C_1 e^t + C_2 e^{-t}.$$

Condición inicial:

$$x^*(0) = 1$$

$$C_1 e^0 + C_2 e^{-0} = 1$$

$$C_1 * 1 + C_2 * 1 = 1$$

$$C_1 + C_2 = 1$$

$$C_2 = 1 - C_1.$$

Condición de transversalidad:

$$F_{x'}|_{t=T} = 0$$

$$2x'(t)|_{t=T} = 0$$

$$2x'(T) = 0$$

$$x'(T) = \frac{0}{2}$$

$$x'(T) = 0$$

$$C_1 e^T - C_2 e^{-T} = 0$$

$$C_1 e^T - (1 - C_1) e^{-T} = 0$$

$$C_1 e^T - e^{-T} + C_1 e^{-T} = 0$$

$$C_1 (e^T + e^{-T}) = e^{-T}$$

$$C_1 = \frac{e^{-T}}{e^T + e^{-T}}$$

$$C_1 = \frac{\frac{1}{e^T}}{e^T + \frac{1}{e^T}}$$

$$C_1 = \frac{\frac{1}{e^T}}{\frac{e^{2T}+1}{e^T}}$$

$$C_1 = \frac{1}{e^{2T}+1}.$$

$$C_2 = 1 - \frac{1}{e^{2T}+1}$$

$$C_2 = \frac{e^{2T}+1-1}{e^{2T}+1}$$

$$C_2 = \frac{e^{2T}}{e^{2T}+1}.$$

$$x^*(t) = \frac{1}{e^{2T}+1} e^t + \frac{e^{2T}}{e^{2T}+1} e^{-t}, \text{ con } t \in [0, T]. \quad \text{Extremal del funcional.}$$

Condición de Legendre (necesaria):

$$F_{x'x'} = 2 > 0.$$

Condición del Hessiano (suficiente):

$$|H F(t, x, x')| = \begin{vmatrix} F_{xx} & F_{xx'} \\ F_{x'x} & F_{x'x'} \end{vmatrix}$$

$$|H F(t, x, x')| = \begin{vmatrix} 2 & 0 \\ 0 & 2 \end{vmatrix},$$

donde:

$$|H_1| = 2 > 0 \text{ y } |H_2| = 4 > 0.$$

Por lo tanto, $x^*(t)$ es el extremal del funcional y realiza el mínimo, ya que la matriz $H F(t, x, x')$ es definida positiva $\forall t, x, x' \in Dom_F$, ya que $|H_j| > 0, j=1, 2$, y, entonces, $F(t, x, x')$ es estrictamente convexa, $\forall t, x, x' \in Dom_F$.

Ejercicio 5.

Encontrar la curva que une los puntos $(0, 0)$ y $(1, 0)$ de modo que la integral

$$\int_0^1 (x'')^2 dt$$

alcanza el valor mínimo, sabiendo que $x'(0) = a$ y $x'(1) = b$.

Se definen las funciones:

$$x_1(t) = x(t).$$

$$x_2(t) = x'(t).$$

$$x_1'(t) = x_2(t).$$

El problema que se quiere resolver puede expresarse de la siguiente manera:

$$\min_{\{x_1, x_2\}} \int_0^1 (x_2')^2 dt, \quad \text{con } x_1 - x_2 = 0; \quad x_1(0) = 0; x_1(1) = 0; x_2(0) = a; x_2(1) = b.$$

$$\mathcal{L}(t, x_1, x_2, x_1', x_2', \lambda, \lambda') = (x_2')^2 + \lambda [x_1' - x_2].$$

Condición de Euler para x_1 :

$$\mathcal{L}_{x_1} - \frac{d}{dt} (\mathcal{L}_{x_1'}) = 0$$

$$0 - \frac{d}{dt} \lambda = 0$$

$$\frac{d}{dt} \lambda = 0$$

$$\lambda(t) = A.$$

Condición de Euler para x_2 :

$$\mathcal{L}_{x_2} - \frac{d}{dt} (\mathcal{L}_{x_2'}) = 0$$

$$-\lambda - \frac{d}{dt} (2x_2') = 0$$

$$-\lambda - 2x_2'' = 0$$

$$2x_2'' = -\lambda$$

$$x_2'' = \frac{-\lambda}{2}$$

$$x_2'' = \frac{-A}{2}$$

$$\int x_2'' dt = \int \frac{-A}{2} dt$$

$$x_2' = \frac{-A}{2} t + B$$

$$\int x_2' dt = \int \left(\frac{-A}{2} t + B \right) dt$$

$$x_2 = \int \frac{-A}{2} t dt + \int B dt$$

$$x_2 = \frac{-A}{2} \int t dt + B \int dt$$

$$x_2 = \frac{-A}{2} \frac{t^2}{2} + Bt + C$$

$$x_2 = \frac{-A}{4} t^2 + Bt + C.$$

$$x_1' = x_2$$

$$x_1' = \frac{-A}{4} t^2 + Bt + C$$

$$\int x_1' dt = \int \left(\frac{-A}{4} t^2 + Bt + C \right) dt$$

$$x_1 = \int \frac{-A}{4} t^2 dt + \int Bt dt + \int C dt$$

$$x_1 = \frac{-A}{4} \int t^2 dt + B \int t dt + C \int dt$$

$$x_1 = \frac{-A}{4} \frac{t^3}{3} + B \frac{t^2}{2} + Ct + D$$

$$x_1 = \frac{-A}{12} t^3 + \frac{B}{2} t^2 + Ct + D.$$

Condiciones iniciales y finales:

$$x_1(0) = 0$$

$$\frac{-A}{12} * 0^3 + \frac{B}{2} * 0^2 + C * 0 + D = 0$$

$$\frac{-A}{12} * 0 + \frac{B}{2} * 0 + 0 + D = 0$$

$$0 + 0 + 0 + D = 0$$

$$D = 0.$$

$$x_1(1) = 0$$

$$\frac{-A}{12} * 1^3 + \frac{B}{2} * 1^2 + C * 1 = 0$$

$$\frac{-A}{12} + \frac{B}{2} + C = 0.$$

(1)

$$x_2(0) = a$$

$$\frac{-A}{4} * 0^2 + B * 0 + C = a$$

$$\frac{-A}{4} * 0 + 0 + C = a$$

$$0 + 0 + C = a$$

$$C = a.$$

$$x_2(1) = b$$

$$\frac{-A}{4} * 1^2 + B * 1 + a = b$$

$$\frac{-A}{4} * 1 + B + a = b$$

$$\frac{-A}{4} + B + a = b$$

$$B = b - a + \frac{A}{4}.$$

(2)

Reemplazando (2) en (1), se tiene:

$$\frac{-A}{12} + \frac{b - a + \frac{A}{4}}{2} + a = 0$$

$$\begin{aligned} \frac{-A}{12} + \frac{b}{2} - \frac{a}{2} + \frac{A}{8} + a &= 0 \\ \frac{A}{24} + \frac{b}{2} + \frac{a}{2} &= 0 \\ \frac{A}{24} &= -\frac{a}{2} - \frac{b}{2} \\ \frac{A}{24} &= \frac{-(a+b)}{2} \\ A &= \frac{-(a+b)}{2} * 24 \\ A &= -12(a+b). \end{aligned}$$

(3)

Y, reemplazando (3) en (2), se tiene:

$$\begin{aligned} B &= b - a - \frac{12(a+b)}{4} \\ B &= b - a - 3(a+b) \\ B &= b - a - 3a - 3b \\ B &= -4a - 2b \\ B &= -2(2a + b). \end{aligned}$$

Por lo tanto, se tiene:

$$\begin{aligned} x^*(t) &= \frac{-[-12(a+b)]}{12} t^3 - \frac{2(2a+b)}{2} t^2 + at \\ x^*(t) &= (a+b) t^3 - (2a+b) t^2 + at, \quad \text{con } t \in [0, 1]. \quad \text{Extremal del funcional.} \end{aligned}$$

Condición de Legendre (necesaria):

$$\begin{aligned} |\mathcal{L}_{x'x'}| &= \begin{vmatrix} \mathcal{L}_{x_1'x_1'} & \mathcal{L}_{x_1'x_2'} \\ \mathcal{L}_{x_2'x_1'} & \mathcal{L}_{x_2'x_2'} \end{vmatrix} \\ |\mathcal{L}_{x'x'}| &= \begin{vmatrix} 0 & 0 \\ 0 & 2 \end{vmatrix} \\ |\mathcal{L}_{x'x'}| &= 0. \end{aligned}$$

Condición del Hessiano (suficiente):

$$\begin{aligned} |H \mathcal{L}(t, x_1, x_2, x_1', x_2', \lambda, \lambda')| &= \begin{vmatrix} \mathcal{L}_{x_1x_1} & \mathcal{L}_{x_1x_2} & \mathcal{L}_{x_1x_1'} & \mathcal{L}_{x_1x_2'} \\ \mathcal{L}_{x_2x_1} & \mathcal{L}_{x_2x_2} & \mathcal{L}_{x_2x_1'} & \mathcal{L}_{x_2x_2'} \\ \mathcal{L}_{x_1'x_1} & \mathcal{L}_{x_1'x_2} & \mathcal{L}_{x_1'x_1'} & \mathcal{L}_{x_1'x_2'} \\ \mathcal{L}_{x_2'x_1} & \mathcal{L}_{x_2'x_2} & \mathcal{L}_{x_2'x_1'} & \mathcal{L}_{x_2'x_2'} \end{vmatrix} \\ |H \mathcal{L}(t, x_1, x_2, x_1', x_2', \lambda, \lambda')| &= \begin{vmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2 \end{vmatrix}, \end{aligned}$$

donde:

$$|H_1| = 0, |H_2| = 0, |H_3| = 0 \text{ y } |H_4| = 0.$$

Por lo tanto, $x^*(t)$ es el extremal del funcional y realiza el mínimo, ya que, a pesar de que la condición del Hessiano no aporta ninguna información, ya que $|H_j| = 0$, $j = 1, 2, 3, 4$, la matriz $H \mathcal{L}(t, x_1, x_2, x'_1, x'_2, \lambda, \lambda')$ es semidefinida positiva $\forall t, x_1, x_2, x'_1, x'_2, \lambda, \lambda' \in Dom_{\mathcal{L}}$, ya que todos los elementos de la diagonal principal son mayores o iguales a 0, y, entonces, $\mathcal{L}(t, x_1, x_2, x'_1, x'_2, \lambda, \lambda')$ es convexa, $\forall t, x_1, x_2, x'_1, x'_2, \lambda, \lambda' \in Dom_{\mathcal{L}}$.